

▣ 09 Counting

Problem 1

- (a) How many distinct 4-digit credit card PINs are possible?
- (b) What if the 4 digits must be one of 1, 2, 3, 4, 5 or 6?
- (c) What if the 4 digits may be in the range 0-9 but at least 2 different digits must be used?
- (d) What if the PIN may not start with a zero?
- (e) What if the PIN may not be odd?
- (f) What if the PIN must not have any repeated digits?

▣ Permutations

The last part is an example of a permutation problem. i.e., we must select a sequence of length k ($k = 4$ here) from a set of n symbols ($n = 10$ here) where no symbol may be selected twice.

The number of possible sequences is given by ${}^n P_r = n! / (n - k)!$ where $n! = (n)(n-1) \dots (2)(1)$

Here we want ${}^{10}P_4 = 10! / 6! = 10 \times 9 \times 8 \times 7 = 5040$ as before.

Permutations are widely used and are appropriate to count say, the number of possible finishes (1^{st} , 2^{nd} , 3^{rd}) in a race, or the number of ways a DJ might select and arrange the play order for 4 tracks from an album of 12.

What if the PIN must have at least some digits repeated? *Exercise*

Problem 2

How many sequences of 6 letters are possible from the lower-case letters a, b, c, ..., z if

- (a) at least two different letters must be used in the sequence
- (b) the first letter may not be x, y or z
- (c) none of the letters may be x, y, or z

Problem 3

- (a) How many different 3-character passwords can be made with the 26 lowercase letters and the 10 digits with no restriction?
- (b) What if the same character may not be repeated 3 times?
- (c) What if the same character may not be repeated at all?
- (d) What if the third character must be a digit?
- (e) What if the third character must be a digit and the first two characters must be unique?
- (f) What if the password must start with a letter and have all characters unique?

▣ Combinations

A common feature of all the examples so far is that order matters. For example, the PIN 6763 is not the same as 6736 or 6673 etc. There are many situations where order does not matter. For example, the numbers selected in the lotto (or bingo etc.), the number of ways a committee of 3 can be selected from a membership of 8 etc. These are example of combinations.

Example How many ways can 3 counties be selected from the 6 in Munster?

We might proceed as before and argue that there are 6 choices for the first selection, 5 for the second and 4 for the third and take the total to be $6 \times 5 \times 4 = 120$. However, there is an element of multiple counting here. For example, the selection Cork (C), Waterford (W) and Limerick (L) has been counted a total of $3! = 6$ times as there are 6 distinct permutations of these counties (CLW, CWL, LCW, LWC, WCL, WLC) but as combinations, these are all the same. In fact, every possible triple of counties has been counted $3!$ times so to correct for this we take $(6 \times 5 \times 4) / 3! = 20$

It can be readily seen than in general the number of ways of selecting combinations of size k items from n is given by $n! / (n-k)!k!$ or nC_k . This (along with permutations) is a function on most calculators.

Thus, the number of ways of selecting 6 numbers from 45 is ${}^{45}C_6 = 8,145,060$. The number of ways 5 cards may be selected from a pack of 52 is ${}^{52}C_5 = 2,598,960$ etc.

Problem 4

100 people each buy one ticket each for a raffle for which there are three prizes.

- (a) How many ways can the prizes be won?
- (b) How many sets of prize winners are there?
- (c) How many ways can the prizes be won if everybody bought at least 3 tickets?

Note that when counting say the number of ways of selecting a team of 11 from a panel of 16, the total will necessarily be the same as the number of ways of selecting the 5 for omission.

Thus ${}^{16}C_{11} = {}^{16}C_5$ or in general ${}^nC_k = {}^nC_{n-k}$.

Problem 5

How many committees of 4 can be formed from 6 men and 8 women?

- (a) with no restriction?
- (b) if all members are to be of the same sex?
- (c) if there is to be 2 of each sex

Note that (b) involved the addition of two nC_r terms while (c) involved their multiplication. This is a common source of error in counting problems with students adding terms they should be multiplying and vice versa.

The difference between them can be seen by noting that (b) uses (implicitly) the conjunction OR as in “committee is to be all male OR all female” while (c) uses (also implicitly) the conjunction AND as in “committee is to have 2 males AND 2 females”

Alternatively, it can be noted that when terms are added as in (b) each term represents a number of ways of selecting complete committees. e.g. each of the 6C_4 ways of forming an all-male committee represents a complete committee of 4.

On the other hand, when terms are multiplied as in (c) then each term represents a number of ways of selecting only part of a committees. i.e., each of the 6C_2 ways of selecting two males for the committee represents only part of the committee, additional terms are needed to enumerate the ways in which the remainder of the committee might be selected.

Problem 6

Playing cards consists of pack of 52 made up of 4 suits hearts, diamonds, clubs and spades with 13 cards in each suit with face values Ace, 2, 3, ..., 9, 10, Jack, Queen and King. There are many variations of poker, we will take a simple version where a player is dealt 5 cards which may not be exchanged. We further assume that there are no wild cards (a wild card permits the holder of such a card to declare its suit or face value to whatever they feel is to their advantage).

- (a) How many ways can a poker hand be dealt?
- (b) How many ways can a royal flush (10, J, Q, K, A of one suit) be obtained?
- (c) How many ways can a straight flush (a run of one suit starting with A, 2, ..., 9) be obtained?
- (d) How many ways can 4 of a kind be obtained?
- (e) How many ways can a house (3 of a kind and 2 of a second kind) be obtained?

- (f) How many flushes are there excluding royal or straight flushes?
- (g) How many straights are there excluding royal or straight flushes?
- (h) How many 3 of a kind are there?
- (i) How many 2 pairs are there?
- (j) How many 1 pair are there?

Classical probability calculation

If 5 cards are selected at random from a pack how likely is it that an ordinary flush will result?

The answer is given by n / N where n is the number of possible ordinary flushes and N is the total number of ways of selecting 5 cards from 52. From above this is $5108 / 2598960 = 0.00197 \approx 0.2\%$

The probabilities of each other hand is tabulated below.

Hand	Frequency	Probability	Cumulative probability
Royal flush	4	0.000154%	0.000154%
Straight flush	36	0.00139%	0.00154%
Four of a kind	624	0.024%	0.026%
Full house	3744	0.144%	0.170%
Flush	5108	0.197%	0.367%
Straight	10200	0.392%	0.76%
Three of a kind	54912	2.1%	2.9%
Two pair	123552	4.8%	7.7%
One pair	1098240	42.3%	50.0%

Note that two pair are likely to be enough to win any game, even with say five players. A pair will be enough more often than not providing there are only two players etc. Of course, most variations of the game of poker are more complex than this. For example, players may be permitted to exchange some of their cards (draw poker) or there may be designated wild cards etc.

Problem 7

A football club have a squad of 28 players consisting of 4 goalkeepers, 9 defenders, 10 midfielders and 5 strikers. The manager opts to play a 4-4-2 formation (i.e., a goalkeeper, 4 defenders, 4 midfielders and 2 strikers). Determine the number of ways a team of 11 players is selected from the squad

- (a) If any player can be selected to play in any position.
- (b) Players may only be selected to play in their designated positions.
- (c) For a training exercise, midfielders must be selected as defenders and vice versa.
- (d) Players play in their designated positions, but two strikers are deemed “too similar” and must not be selected together.
- (e) If two teams are to be selected for a training game, with players playing in position.

Problem 8

A museum curator must assemble a display of 12 archaeological artefacts, of which 5 are pottery pieces, 3 are weapons and 4 are farming tools. The items will be arranged in a linear fashion on a display table. How many ways can the artefacts be arranged if

- (a) There is no restriction
- (b) Similar artefacts must be together
- (c) Three particular pottery pieces and two particular farming tools must be together.
- (d) The weapons must be all arranged together
- (e) No weapons must be arranged together

Solution

(a) $12!$

(b) $5! \cdot 3! \cdot 4! \cdot 3!$ (| pottery|! x | weapons|! x| farm tools|! x| groups|!)

(c) As three pottery pieces (say 1, 2 and 4) and two farm tools (say 2 and 4) are together, there are 8 different “objects” to position

P1P4P2	P3	F2F4	F1	F3	W1	W2	W3
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And for each of these there are $3!$ ways to arrange P1, P4 and P2 and $2!$ ways to arrange F2 and F4

Total = $8! \cdot 3! \cdot 2!$

(d) $10! \cdot 3!$ (Similarly approach to (c))

(e) Note that sometimes a counting problem needed to be partitioned in to a number of sub-calculations. In such cases, care needs to be taken to ensure that there isn’t multiple counting of the same arrangement.

No two weapons together = # no restriction - # with at least two together

We partition “at least 2” in to “exactly 2” or “exactly 3”

= # no restriction - # with exactly two together - # with exactly three together

= $12!$ - # with exactly two together - $10! \cdot 3!$ (from earlier calculations)

Thus, we need only calculate # with exactly two together – and to do this, we further partition the count.

with exactly two together = A + B + C where

A is the number of arrangements beginning with two weapons

B is the number of arrangements ending with two weapons

C is the number of arrangements with two weapons somewhere in the middle.

We need not distinguish between pottery and farm tools artefacts. We represent these with a 1 and the weapons with a 0.

A? E.g., 0 0 1 1 1 1 0 1 1 1 1 1 1

$3!$ Number of permutation of the three weapons

9 Number of objects to go in position 3 (note that the third weapon cannot go here)

$9!$ Number of objects to go in positions 4 to 12

Total = $3! \times 9 \times 9!$

B? This yields the same total as A = $3! \times 9 \times 9!$

C? E.g., 1 1 1 0 0 1 0 1 1 1 1 1 1

$3!$ Number of permutation of the three weapons

9 Number of objects on the left of the two weapons (note that the third weapon cannot go here)

8 Number of objects on the right of the two weapons (note that the third weapon cannot go here either)

$8!$ Number of objects to go in positions 4 to 12

Thus, the total number of permutations with at least two weapons together is

$10! \cdot 3! + 3! \times 9 \times 9! + 3! \times 9 \times 9! + 3! \times 9 \times 8 \times 8!$

Exercise

Show that for the general problem where N objects in total are arranged with at least two of three particular objects are to be adjacent is given by $3!(N-3)!(N-2)^2$

If $N = 12$ then $3!(N-3)!(N-2)^2 = 217,728,000$

So, the total number of ways with no two of the three adjacent = $12! - 217,728,000 = 261,273,600$

And if the arrangement is done at random, the probability that no two of the three will be adjacent is given by

$$261,273,600 / 479,001,600 = 54 / 99 \sim 0.54$$

Note this last computation illustrates that counting (and related probability problems) can be quite tedious.

In such cases, you might elect to devise a simulation, either as an alternative to the computation above, or to verify that the calculations are correct.

A sample simulation is shown below. Interpreting the details is left as an exercise.

If our calculated probability of 0.54 is correct, then repeated runs of the simulation will sometime yield a probability estimate higher than 0.54, and sometimes lower.

```
# Represent weapons with 0 and non-weapons with 1
artefacts = c(0,0,0,1,1,1,1,1,1,1,1,1)

# In 1000 random permutations, we total the number of times
# the weapons are apart in the following variable
apartCount = 0

for(i in 1:1000)
{
  # Generate a random permutation of artefacts - E.g., 1 0 1 1 0 1 0 0 1 1 1 1
  # (This is an integer sequence of 9 1s and 3 0s)
  simN = sample(artefacts)

  # Convert to string to facilitate string processing functions
  # E.g., "101101001111"
  simS = paste(sample(artefacts),collapse="")

  # Look for "00" in the string - i.e., two weapons together
  # (grep returns integer(0) if no match to "00" is found)
  weaponsTogether = grep("00",simS)

  # increment apartCount if "00" not found
  # (identical(A, B) is considered 'safer' than A == B)
  apartCount = ifelse(identical(weaponsTogether,integer(0)),apartCount+1, apartCount)
}
#estimate the probability
p = apartCount/total
```

Problem 9

A goods train must tow 8 carriages, 3 of which are transporting highly flammable goods. For safety reasons, these 3 carriages should be separated by at least one of the other 5 carriages. If the positions of the carriages behind the train are selected at random, how likely is it that this safety requirement will be observed.

Solution

A moment's reflection should convince you that this is essentially the same problem as the previous one involving the artefacts. Hence, the approach above can be adapted to the problem here. (*Exercise*)

However, to demonstrate that these problems can often be tackled in multiple ways, an alternative approach to the solution is considered.

The required probability is given by $1 - \frac{n}{N}$ where n is the number of arrangements of the 8 carriages with at least 2 of the flammable ones adjacent and N is the total number of arrangements of the 8 carriages. N can readily be calculated to be $8!$. The challenge is to calculate n .

Denote the “flammable” carriages $f1$, $f2$ and $f3$ and the “regular” carriages $r1$, $r2$, $r3$, $r4$ and $r5$.

We can partition the count in to three sub-counts. The first, where $f1$ and $f2$ are adjacent, the second where $f1$ and $f3$ are adjacent and the third where $f2$ and $f3$ are adjacent. (The three counts are clearly all the same.)

Consider the number of ways that $f1$ and $f2$ can be adjacent. We in effect have 7 rather than 8 objects to position – an example is shown below.

r3	r1	f1f2	r4	f3	r5	r2
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There are $7!$ ways of arranging these “placeholders” and then $2!$ ways of arranging $f1$ and $f2$ ($f1f2$ or $f2f1$), giving a total of $7! \times 2!$

Note that this count includes the arrangements where $f1$, $f2$ and $f3$ are all together. (i.e., if $f3$ either immediately precedes or follows $f1f2$)

Also note that there are 3C_2 ways to choose 2 from the 3 “flammable” carriages - overall we have ${}^3C_2 \times 7! \times 2!$

However, there is some multiple counting here when $f1$, $f2$ and $f3$ are together.

For example, $r2 r1 f2 f1 f3 r5 r3$ is counted in the $f2f1$ count (with $f3$ immediately after $f1$) – but it is also in the $f1f3$ count (with $f2$ immediately before $f1$). In fact, the arrangements with $f1$, $f2$ and $f3$ together are all counted twice. Thus, we must determine such arrangements there are with $f1$, $f2$ and $f3$ together and subtract.

Using the logic of an earlier example, if $f1$, $f2$ and $f3$ are together, then we in effect have 6 objects rather than 8.

It is left as an exercise to show that this count is $6! \times 3!$ (Hint: refer to the example below)

r4	r1	f3f1f2	r3	r2	r5
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Thus, $n = {}^3C_2 \times 7! \times 2! - 6! \times 3! = 25,920$, $N = 40,320$ and so the required probability is 0.357

Exercise

What if 5 of the 8 carriages are flammable? (Hint: Sometimes a count is very low, it can be obtained by directly enumerating all of the possibilities.)

Exercise

In problem 8, determine the number of arrangements if none of the farm tools may be together

Problem 10

A development team of 9 engineers is to be assembled to develop a new product. The team is selected from three engineering disciplines - there are 12 mechanical, 15 industrial and 17 electrical engineers. Determine the number of ways the team can be assembled if

- There is no restriction.
- The engineers on the team must be all from one discipline.
- There must not be any electrical engineers.
- There must be at least one engineer from each discipline.
- There must be at least two engineers from each discipline.

Solution

Parts (i) to (iv) are not dissimilar from problems considered earlier and are left as an exercise.

To tackle (v), we devise a way to partition the count.

Consider one possible make-up of the team in terms of engineering disciplines – say 4 mechanical, 2 industrial and 3 electrical engineers – (4, 2, 3). The number of such possible teams will be given by $^{12}C_4 \cdot ^{15}C_2 \cdot ^{17}C_3$

There is a similar term for every other possible make-up. To include all of these we must identify every way to add three integer values, each at least 2, and yielding a total of 9 – (3, 3, 3), (2, 2, 5) etc.

It is left as exercise to show that there are 10 such possible combinations. However, rather than manually try to identify these (and run the risk) of omitting some of them, we will use R code to identify these, and also calculate the total required count

It is left as an exercise to show that the following R code yields the required count, and this totals ~ 444 million. Note that this code is *not* a simulation – rather an exact count of all possibilities.

```
splits = c(2, 3, 4, 5)
mecN = 12
indN = 15
eleN = 17

countEnum = 0
for(mec in splits)
  for(ind in splits)
    for(ele in splits)
      if(mec + ind + ele == 9)
        countEnum = countEnum +
          choose(mecN, mec) * choose(indN, ind) * choose(eleN, ele)
print(countEnum)
```

Exercise

1. If the team were selected at random, and it was required to estimate the likelihood of there being at least two engineers from each discipline, then the total above should be divided by the total found in (i). Show that this will yield a probability of about 0.626.
2. Develop a simulation similar to the one presented earlier that yields a probability estimate consistent with the value of 0.626 obtained above.